# PPO-CF CVPR 2027 submission — issues to fix

Nov 1, 2026 · @Sadok Ben Yahia

## Overview

There are 52 issues in six groups. Groups A–D change what the paper claims or can defend at CVPR, and should be settled first. Each item gives its location in the current PDF (Ayoub et al., v3) and a concrete fix. Put your name next to an item when you take it, and tick it when it's done.

| Milestone | Date (AoE) | What it needs from this list |
|---|---|---|
| OpenReview enrollment | Nov 10, 2026 | Final author list (frozen after this), all co-authors with complete OpenReview profiles, and a working title and abstract |
| Paper submission | Nov 16, 2026 | Groups A–C and F done. Title and abstract can still be edited until this date |
| Supplementary material | Nov 23, 2026 | Every "Supp. Sec./Table" referenced in the paper exists (E11) |

Order of work: start the group D runs immediately, because each PPO run takes about 9–11 h per direction (§5.1). Fix group A in parallel, since the abstract numbers depend on it.

## A. Numerical inconsistencies (blocking)

The likely cause of A1–A4 is that the paper mixes two evaluations of the default p→n arm. One was run with the n→p adapter, the other with the p→n adapter (§3.4 and the Table 2 caption). Decide which evaluation is the reference, then update every number below from it.

- [ ] **A1. Mean CV 0.838 is not the mean of Table 2.** Table 2 gives 0.960 (n→p) and 0.670 (p→n), which average to 0.815. 0.838 implies p→n ≈ 0.716, probably the value before re-evaluation. *Where:* abstract; Table 2 "mean" row; §5.3 ("0.838 vs. 0.940"); §5.5 ("0.838 → 0.017"). *Fix:* recompute from the reference evaluation and update all four places.
- [ ] **A2. The nih p→n transfer ratio has two values.** The abstract gives 0.165; Table 3 and the §5.3 threshold text give 0.186 ("crosses at 0.20", "holds only for thresholds up to 0.186"). *Fix:* use one value everywhere and redo the §5.3 threshold counts and wording.

- [ ] **A3. The default p→n mean transfer has two values.** Tables 2 and 4 give 0.276. Table 3's ensemble row averages to (0.186 + 0.308 + 0.384)/3 = 0.293, and §5.5 says "falls from 0.293 to 0.212". *Fix:* same as A1.
- [ ] **A4. Bootstrap CIs don't say which evaluation they come from.** The §5.3 interval for the ensemble's p→n, [0.155, 0.216], contains both 0.165 and 0.186. *Fix:* recompute on the reference evaluation and say which one it is.
- [ ] **A5. The no-policy LHR baseline contradicts itself.** §5.2 gives 2.046 (rsna-only guidance, single-classifier groups) and says the policy's contribution under the default configuration is "unmeasured". The Fig. 1 caption gives 2.064 "with the committed guidance and groups". 2.046 vs 2.064 also looks like swapped digits. *Fix:* check the logs. If 2.064 under the default configuration is real, §5.2 must report the policy's contribution as measured (2.595 vs 2.064, with a paired test), and the abstract can cite it (C7).
- [ ] **A6. Paired comparisons against the default use different adapters.** The default p→n now uses its own adapter, while the single-classifier, 3-model and sparsity arms use the n→p adapter for p→n (§6). So the "LHR vs. default (paired)" column in Table 4 compares across adapters. *Fix:* compare against the default evaluated with the same adapter as the other arm, and say so in the Table 4 caption.
- [ ] **A7. Rounding check.** Table 4 gives the single-classifier p→n transfer as 0.146, but Table 3's row averages to 0.145. *Fix:* check the unrounded value.
- [ ] **A8. Policy-behaviour statistics come from training, not evaluation.** The Introduction, the Table 1 caption and §6 say the default n→p policy holds a7 at 1.298, uses three passes and runs one to two steps. §6 says these come from 20 sampled training episodes, while evaluation uses the deterministic mean action (§4.1). *Fix:* log the actions during evaluation, report those statistics, and update all three places. This is cheap and supports the "near-fixed schedule" claim.

## B. Title

The current title is "Reinforcement-Learning-Scheduled Latent Diffusion for Visual Counterfactual Explanations, and What Ground-Truth-Anchored Evaluation Reveals About Them". Replacement options are in the last section.

- [ ] **B1. It names the wrong evaluation.** The finding that undercuts CV comes from held-out transfer, which is anchored to independent classifiers, not to ground truth. The ground-truth metric (LHR) gives the *favourable* result (2.60×). The abstract's last sentence and the Conclusion correctly say "ground-truth- or independence-anchored".

- [ ] **B2. It claims more than the evidence supports.** "…Reveals About Them" sounds like a finding about visual counterfactual explanations in general. The evidence is one system, one dataset and one run per arm.
- [ ] **B3. The domain is missing.** Every result is on chest radiographs (RSNA Pneumonia); the title should say so.
- [ ] **B4. It puts the method first.** The abstract calls the evaluation "the more consequential" contribution. A method-first title invites a method review, which the paper is not equipped to win (see D1, D4).
- [ ] **B5. It is too long and awkward.** 21 words, a triple-hyphen compound, and "Them" left alone on its own line in the PDF.

## C. Abstract

The abstract has to be in OpenReview by 10 Nov and can still be edited until 16 Nov. A rewrite that addresses C1–C13 is in the last section.

- [ ] **C1. "Two contributions" vs three.** The abstract says two; the §1 contribution list has three. "Two contributions that pull in opposite directions" also reads as self-undermining to CVPR reviewers. *Fix:* present one thesis, with PPO-CF as the case it is tested on.
- [ ] **C2. It oversells the RL scheduler.** The abstract presents it as per-image adaptivity. §6 says the learned schedule is close to fixed, there's no tuned fixed-schedule baseline, and trained vs random (d = 0.189) covers only the single-classifier policy. *Fix:* say in one clause that the learned schedule is close to a fixed, aggressive one.
- [ ] **C3. Muddled logic.** "Under these, the picture changes substantially" is followed by the favourable LHR. What changes the picture is transfer and how diffuse the edits are. *Fix:* restructure so that LHR is the positive result, and transfer plus diffuseness are the problem.
- [ ] **C4. Diffuse edits are missing.** About 65% of pixels change, and only 25–31% of the change sits in the top 5% of pixels (Table 2). This is a headline in §5.2 and §7. *Fix:* add one clause.
- [ ] **C5. It pairs numbers from different directions.** A two-direction mean CV (0.838) is set against an n→p transfer ratio. *Fix:* compare within n→p: CV 0.960 vs nih transfer 0.046.
- [ ] **C6. The "central claim" uses different numbers in different places.** §5.3 calls the single-classifier pair (CV 0.980, transfer 0.080) "the paper's central empirical claim", while the abstract uses the ensemble's numbers. *Fix:* use the same numbers in both, preferably the default (ensemble) policy's or both policies'.
- [ ] **C7. The no-policy baseline is missing.** 2.60× LHR lift is reported without the about 2.05–2.06× reached without a policy. *Fix:* add it once A5 is resolved.

- [ ] **C8. "We introduce two measurements" claims too much novelty.** See E1 and E2. *Fix:* use "adapt" or "pair CV with".
- [ ] **C9. "Largely confined to a dilated lung region" isn't backed by a number.** See E4. *Fix:* soften, or add the in-mask share once E4 is done.
- [ ] **C10. The ensemble vs single-classifier sentence carries three caveats.** Single run, different images, different hardware — together they signal weak evidence. *Fix:* either re-run it (D3) and keep it, or move it out of the abstract.
- [ ] **C11. Internal jargon and undefined terms.** "Committed configuration", n→p and p→n, "most independent validator", "adversarial regime". *Fix:* say "default", spell out the directions, and define or drop the rest.
- [ ] **C12. A section reference and a p-value.** *Fix:* remove "(§4.4)" and "$p = 6.6 \times 10^{-53}$".
- [ ] **C13. Too long.** About 350 words, spilling into the right column. CVPR has no hard limit, but 150–250 words is the norm. *Fix:* aim for 250 words or fewer.

## D. Positioning and missing experiments

As written, the paper is a single-system case study presented as either a method paper or a finding about the whole field. CVPR reviewers will judge it against both. D1 and D3 are the most valuable additions before 16 Nov. The rest are cheap or need only text.

- [ ] **D1. No head-to-head baseline.** §2 says "we do not compare head-to-head (Supp. E3)", which is the most likely reason for rejection if the paper is read as a method paper. *Fix:* compute CV, LHR and transfer for at least one public baseline (DiME [7] or ACE [8]) on the same split and images. This also turns the evaluation argument into a claim about the field rather than about PPO-CF alone.
- [ ] **D2. FiRe [20] counts as prior work, not contemporaneous.** It is arXiv 2608.08664 (August 2026), before the CVPR 2027 cutoff of 15 Sep 2026 (CVPR 2027 Call for Papers). It hand-designs adaptive guidance and early stopping, which are the controls PPO-CF learns. *Fix:* compare against it if feasible; at minimum, discuss the difference explicitly in §1 and §2.
- [ ] **D3. Single run per arm, with confounds.** The single-classifier and ensemble runs differ in evaluation images (78 of 200 p→n sources shared) and hardware. *Fix:* re-run both arms with at least 3 PPO seeds, on the same hardware and the same evaluation images (fix X_n/X_p with one common selector). This would let the opposite rankings from CV and the anchored metrics lead the abstract.
- [ ] **D4. No tuned fixed-schedule baseline.** The learned policy is close to fixed (A8). *Fix:* evaluate the fixed schedule that matches the policy's mean actions (a7 = 1.30, three passes, one to two steps), plus a small grid of fixed schedules. No training is needed. If a fixed schedule matches PPO-CF, say so: the contribution is then the evaluation.

- [ ] **D5. Claims about the field rest on one system.** §1 calls this "a reporting problem for the subfield"; the abstract says CV "can substantially overstate progress". *Fix:* support them with D1, or narrow them to "our system".
- [ ] **D6. Action ranges may limit the policy.** a7 sits at its upper bound (1.298 of 1.30), and the a1 range was narrowed (Table 1 †). Reviewers will ask whether a wider range would change the behaviour. *Fix:* address it in one sentence; optionally run once with a wider a7 range.
- [ ] **D7. Single dataset.** Only RSNA Pneumonia is used, and §6 doesn't list this as a limitation. *Fix:* add it to §6.
- [ ] **D8. Guidance classifier may overlap with the evaluation data.** The rsna classifier was developed on RSNA data (§6), which can inflate CV and possibly LHR. *Fix:* state which validators have no RSNA exposure. This is also the justification E5 needs.

## E. Body: claims, clarity and related work

- [ ] **E1. LHR is close to an existing metric.** It is essentially the energy-based pointing game (Wang et al., Score-CAM, CVPRW 2020) applied to $|x_c - x|$ and normalised by box area. *Fix:* cite it, say "adapt" rather than "introduce", and state what is new (the difference map, the area-null lift, the calibration in §4.2).
- [ ] **E2. Held-out transfer has precedents.** Re-scoring with a classifier not used for generation is close to the oracle-classifier metrics in DiME [7] and ACE [8] and to robust-classifier checks in DVCE [1]. *Fix:* cite them and present the novelty as the pairing, the directional ratio and the threshold analysis.
- [ ] **E3. A related-work sentence is too broad.** "None reports LHR or held-out transfer" (§2) only covers ECED, FiRe and DocVCE. *Fix:* limit it to those methods, or check it against the wider literature.
- [ ] **E4. Confinement to the lungs isn't measured.** About 65% of pixels change by more than 0.01, which probably includes VAE reconstruction error across the whole image. *Fix:* report the share of $|x_c - x|$ that falls inside the dilated mask. Also report the fraction of pixels changed relative to the VAE reconstruction $D(E(x))$, not only relative to x.
- [ ] **E5. "Most independent validator" is never defined.** nih carries the headline numbers (abstract, §5.3, §6, §7), but the main text never says why it is the most independent. *Fix:* define it in §4.3 by its training data and its overlap with the guidance models (see D8).
- [ ] **E6. The novelty claim for RL-scheduled sampling needs checking.** §2 calls it "comparatively unexplored". *Fix:* check against work on learned or adaptive sampling and guidance schedules (e.g., Align Your Steps, Sabour et al. 2024; guidance-weight schedulers, Wang et al. 2024), then cite it or soften the claim.

- [ ] **E7. References are thin (20 entries) and miss expected work on chest X-ray counterfactuals and editing.** Candidates: Gifsplanation via Latent Shift (Cohen et al., MIDL 2021, same group as TorchXRayVision), RadEdit (Pérez-García et al., ECCV 2024), LDCE (Farid et al., 2023), FastDiME (Weng et al., ECCV 2024). *Fix:* verify the venues, add them and position the paper against them.
- [ ] **E8. FID is unreliable at n ≈ 200.** FID uses a 1024-d feature covariance, which can't be estimated well from 200 samples; §6 already admits high variance. *Fix:* add KID or bootstrap CIs, or drop FID from the main table.
- [ ] **E9. The HF-corr definition is ambiguous.** "Pearson correlation of high-pass-filtered images rescaled to [0,1] (0.5 means uncorrelated)" doesn't say whether the images or the correlation are rescaled. *Fix:* write it as (r + 1)/2 if that is what is meant.
- [ ] **E10. 78 vs 76 shared images.** §4.4 and §5.3 say 78 of 200 p→n sources are shared; §5.4 uses 76 shared annotated images. *Fix:* add a clause explaining the two-image difference.
- [ ] **E11. The main text leans heavily on the supplement.** It references Supp. Sec. R1–R6, E1–E3 and Tables R1–R4, and CVPR reviewers are not obliged to read the supplement. *Fix:* make sure the definitions behind each claim (LHR details, the bootstrap, validator provenance) are in the main text. Check that every referenced supplement item exists by 23 Nov.

## F. Formatting and CVPR 2027 compliance

Rules are from the CVPR 2027 Author Guidelines.

- [ ] **F1. Wrong template year.** The PDF header still says "CVPR 2026 Submission". *Fix:* move to the CVPR 2027 template.
- [ ] **F2. LLM Usage Statement is missing.** It is required after the main text and before the references. It doesn't count toward the page limit and must not reveal the authors. It names the tools used, which parts of the work they affected, and how much; if none were used, it says so.
- [ ] **F3. Page budget.** The limit is 8 pages including figures and tables. The text currently ends on page 8. *Fix:* when adding D1, D4 or the E-items, cut elsewhere (e.g., §5.6, or move detail from §5.4 to the supplement).
- [ ] **F4. Reference capitalisation.** BibTeX has lowercased acronyms: [4] Torchxrayvision → TorchXRayVision; [5] gans → GANs; [6] Lora → LoRA; [9] Repaint → RePaint; [11] Sdedit → SDEdit; [14] Rsna → RSNA; [17] Grad-cam → Grad-CAM. *Fix:* protect them with braces in the .bib file.
- [ ] **F5. Title layout.** Check that the new title doesn't leave a word alone on its own line.

- [ ] **F6. Anonymity.** The PDF metadata has no author name. *Fix:* also check the supplement, any code archive and any links for identifying information. External links that add content are not allowed.
- [ ] **F7. OpenReview logistics.** All co-authors need complete OpenReview profiles by 10 Nov. Authors can't be added or removed after that date, only reordered. Check the reciprocal-reviewer requirement: if no author is an eligible reviewer, each author may be on only one submission.

## Proposed title and abstract

Both lead with the evaluation and treat PPO-CF as the case study (B1–B5, C1–C13). Use the first title unless D1 adds a public baseline, which would support a broader claim.

```
\title{High Validity, Low Transfer: Anchored Evaluation of Diffusion
Counterfactual Explanations for Chest Radiographs}
% alternative:
\title{Counterfactual Validity Is Not Enough: Anchored Evaluation of
Diffusion Counterfactuals for Chest X-Rays}
```

The abstract is about 240 words. Lines marked VERIFY or KEEP ONLY depend on A5 and D3.

```
\begin{abstract}
Visual counterfactual explanations are conventionally judged by
Counterfactual Validity (CV): the rate at which an edit flips the decision of
the classifier that guided it. Because one model both steers and scores the
edit, CV cannot separate a class-relevant change from an adversarial
perturbation. We study this circularity on chest radiographs with PPO-CF, a
latent diffusion counterfactual generator in which a PPO agent sets the
sampling schedule (noise strength, guidance scales, refinement passes,
stopping) around a frozen, LoRA-adapted backbone, and a soft background lock
largely confines edits to the lungs. We pair CV with two anchored
measurements: a Lesion Hit Rate (LHR) against radiologist-drawn opacity
boxes, and a held-out transfer ratio that re-scores counterfactuals with
classifiers never used for guidance. On RSNA Pneumonia, normal-to-pneumonia
counterfactuals reach 96\% CV, yet the most independent held-out classifier
reproduces only 4.6\% of the guidance shift. Pneumonia-to-normal edits do
concentrate on annotated opacities (2.60$\times$ LHR lift over the area null;
2.06$\times$ without a learned policy), % VERIFY (A5)
but every configuration we train produces spatially diffuse edits. Two
mechanisms designed to localize edits, an annealed sparsity reward and a
Grad-CAM constraint, both reduce lesion targeting, the latter collapsing
validity to 1.7\%. CV and the anchored metrics also rank our two guidance
variants in opposite order. % KEEP ONLY if D3 confirms
Judged by CV alone, our own system would appear far more successful than it
is; we argue that anchored companion metrics should accompany CV whenever it
is computed with the guidance classifier.
\end{abstract}
```

If D1 adds a public baseline, change the last sentence to a claim about the field.

## Sources

- CVPR 2027 Call for Papers — dates and the contemporaneous-work cutoff
- CVPR 2027 Author Guidelines — page limit, template, LLM Usage Statement, OpenReview rules